



BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

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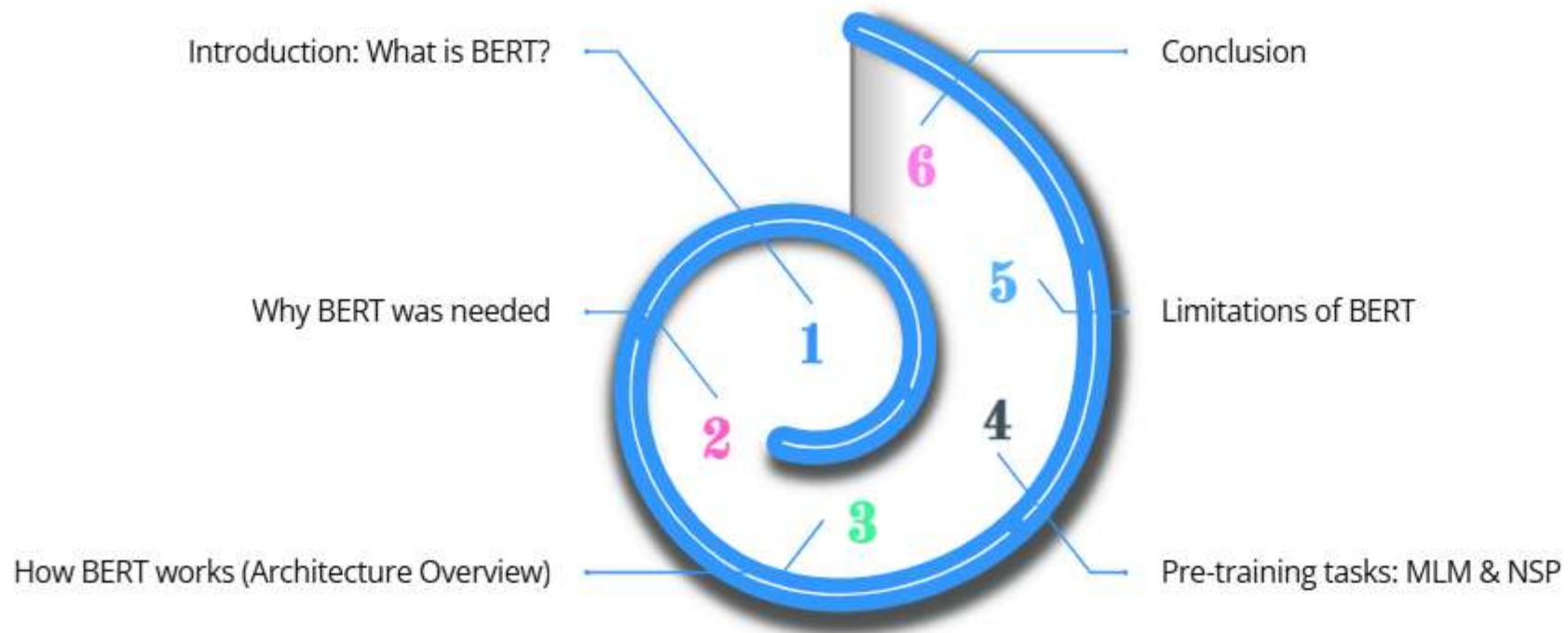
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Agenda: What We'll Cover About BERT

We'll talk about 6 points



Introduction

What is BERT.



- 1** BERT = Bidirectional Encoder Representations from Transformers
- 2** Developed by Google AI Language (2018)
- 3** A deep bidirectional transformer encoder trained on unlabeled text
- 4** Learns meaning using context from both left and right
- 5** Breakthrough model for NLP understanding
- 6** Achieved state-of-the-art performance on 11 NLP benchmarks



Why BERT

How deep bidirectional pre-training fixes unidirectional and shallow approaches for broad NLP tasks

- Earlier models (GPT, LSTMs, ELMo) were unidirectional or shallow bidirectional
- Unidirectional models lose right-side context, hurting understanding
- Many NLP tasks need sentence relationships, not isolated tokens
- Needed a general-purpose pre-trained model that works for multiple tasks
- BERT fixes this using true deep bidirectional attention
- Eliminates the need for heavy task-specific architectures



Architecture

Concise breakdown of layers, embeddings, attention, and special tokens for practical understanding

1 Based on multi-layer Transformer Encoder

- BERT Base → 12 layers, 110M parameters
- BERT Large → 24 layers, 340M parameters
- Uses multi-head self-attention to learn relationships between all words



2 Input embeddings include

- Token embeddings
- Segment embeddings (Sentence A/B)
- Position embeddings



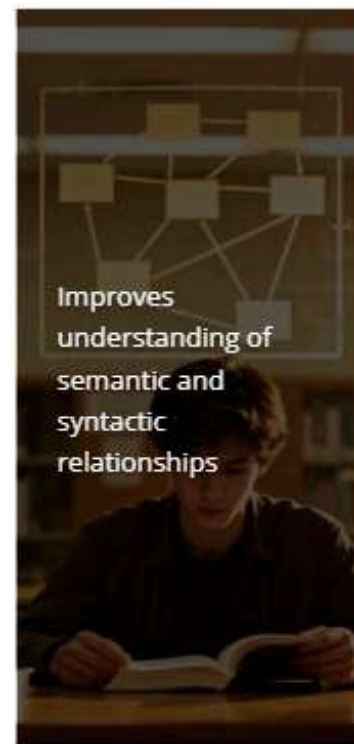
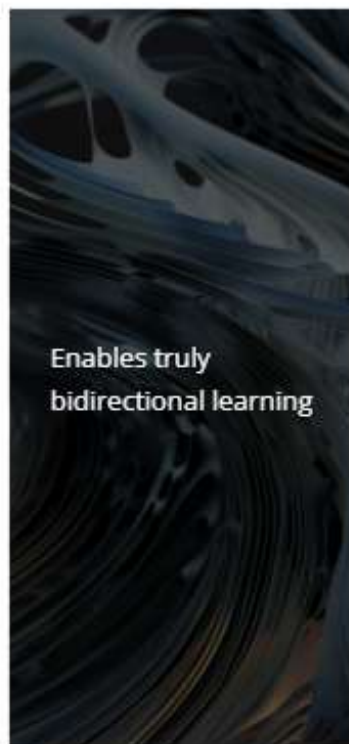
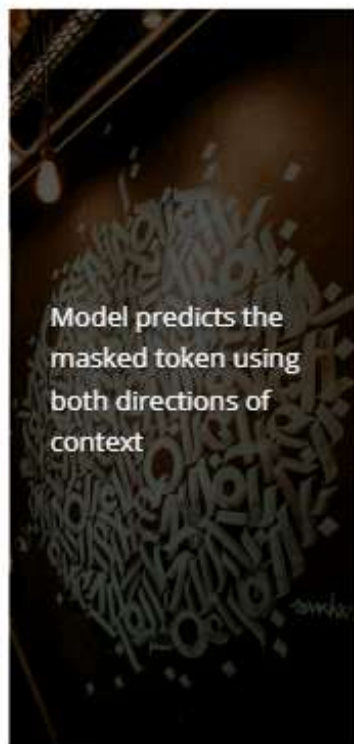
3 Special tokens

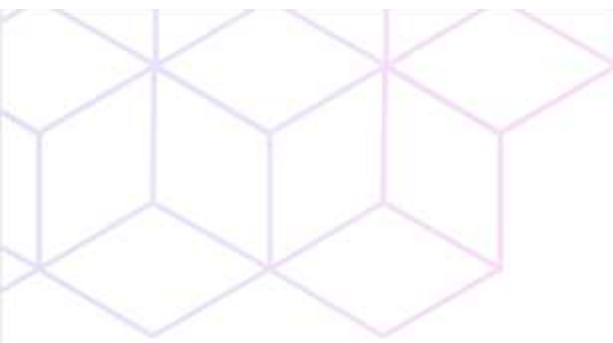
- CLS for classification
- SEP for separation



Masked Language Modeling: Bidirectional Pre-Training That Teaches Context

How MLM masks tokens, predicts them from both sides, and strengthens semantic and syntactic understanding





Next Sentence Prediction (NSP)

How BERT learns sentence relationships using a 50/50 training split and real examples



Task: Determine whether Sentence B follows Sentence A

- Task: Determine whether Sentence B follows Sentence A
- Training setup: 50% → Actual next sentence
- Training setup: 50% → Random sentence
- Helps BERT understand: Sentence relationships



Benefits: Helps BERT understand sentence relationships and reasoning

- Helps BERT understand: Logical continuity
- Helps BERT understand: Reasoning in Q&A and NLI tasks
- Example: A: "He went to the store." B: "He bought milk." → IsNext



Pre-training Data, Sequence Length & Compute

Clear summary of dataset size, training schedule, and
compute resources



Trained on two massive corpora: BooksCorpus (800M words) and English Wikipedia (2.5B words)



Total ~3.3 billion words



Trained for ~1 million steps



Sequence length pre-training: First 90% with length 128



Sequence length pre-training: Last 10% with length 512



Uses huge computational resources (TPUs)



Limitations

Key constraints that impact deployment, performance, and real-time use

Very high computational cost for training

Large model sizes → slow inference

Maximum sequence length = 512 tokens

NSP task later found to be suboptimal (replaced in newer models)

Not efficient for real-time applications

Requires large memory and GPU/TPU power

Struggles with very long documents and knowledge-intensive tasks



Conclusion: Impact on AI and NLP

How BERT changed language understanding, inspired successors, and set new performance standards



BERT transformed NLP with deep bidirectional understanding



Solved major limitations of earlier unidirectional models



Introduced effective pre-training tasks: **MLM + NSP**



Achieved breakthrough performance across major benchmarks



Inspired next-generation models: RoBERTa, ALBERT, DistilBERT, and modern LLMs



Remains one of the most influential architectures in AI





Thank You

